

Adnan Mashrur Sadad

AI Agent Algorithm Engineer — Global E-Commerce Affiliate

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Summary

Built and shipped an LLM agent runtime end to end — a self-correcting repair loop, adaptive follow-up reasoning, and a mocked, CI-gated regression harness at 87% coverage — with no post-training budget, no team, and no one to blame if it broke. Looking for the team and the scale to find out what the same instincts do with real traffic and real evaluation data.

Technical Skills

- **GenAI / Agentic Systems:** Agent Runtime Design, Agentic Memory & Context Engineering, Tool-Use / Tool Routing, RAG, Text2SQL, LLM-as-Judge Evaluation, Structured Output Validation & Repair Loops, Regression Harness Design, Prompt Engineering, Groq API, Whisper
- **Machine Learning & Data:** XGBoost, Scikit-learn, TensorFlow, Pandas, NumPy, Hugging Face
- **Languages:** Python, Java, C++, JavaScript, TypeScript, SQL
- **Frameworks & Deployment:** FastAPI, React, SQLAlchemy, Docker, REST APIs, Microservices, CI/CD (GitHub Actions), OpenAPI

Projects

InterviewPilot — AI Agent Runtime for Structured Interview Evaluation 2026 – GitHub

Python, FastAPI, SQLAlchemy, React/TypeScript, Groq API (Whisper-large-v3, Llama 3.3 70B), Docker, GitHub Actions

- **Closed the gap between raw LLM output and a machine-readable contract with zero dropped requests**, by building a self-correcting repair loop that re-validates and re-prompts Llama 3.3 70B against a JSON schema until the response is trustworthy.
- **Replaced a fixed interview script with adaptive, context-aware questioning**, by building a follow-up agent that reasons over the candidate's prior answers each turn to decide whether to probe deeper or move on.
- **Kept agent and evaluator behavior verifiable without live inference cost**, by building a 17-test pytest regression harness at 87% backend coverage with Groq calls fully mocked, running in GitHub Actions on every push.

Mindhive Chatbot — Multi-Turn Tool-Routing Agent

2025 – GitHub

FastAPI, RAG, Text2SQL

- **Held a coherent multi-turn conversation across 3–5 related turns**, by designing stateful, intent-based planning that resolves each new message against prior context.
- **Proved retrieval quality before shipping**, by building a hand-labeled 20-query offline evaluation harness that scored 85% Hit Rate@3 overall (100% on outlet lookups), MRR 0.86.
- **Routed user intent to the correct tool with zero SQL-injection exposure**, by building two FastAPI microservices — a RAG endpoint validated against 200+ product documents and a Text2SQL endpoint with query sanitization — both documented via OpenAPI.

AI Financial Fraud Detection System

2026

Python, XGBoost, Scikit-learn, FastAPI, Streamlit

- **Caught fraud at a 0.17% base rate, where a model that always predicts “not fraud” scores 99.8% accuracy and 0% recall**, by choosing PR-AUC (0.89) over accuracy as the headline metric and combining SMOTE with class-weighted loss.
- **Reached 0.98 ROC-AUC / 0.89 PR-AUC on held-out data across 284,807 transactions**, by ensembling Random Forest, XGBoost, and unsupervised Isolation Forest so the system covers both known fraud patterns and unlabeled anomalies.

Work Experience

Joget Inc.

Oct 2024 – Mar 2025

Software Developer Intern (Hybrid)

Kuala Lumpur, MY

- **Cut manual processing effort by 30% on a core client workflow**, by integrating REST APIs between Joget DX and external services to automate a previously manual data flow.
- **Shipped production features inside real sprint cycles**, by building and deploying enterprise workflow automations and UI components on the Joget DX low-code platform, from design through CI/CD-managed release.

Education

Universiti Teknologi Malaysia (UTM), MJIT

2021 – 2026

Bachelor of Software Engineering, CGPA: 3.50/4.00 — Dean's List 2022/2023 — Top Performer, Computational Intelligence (2023)*

Kuala Lumpur, Malaysia

*Final CGPA pending official publication.

Achievements

- Gold Medal & Best Video Award, myHCI-UX Student Design Challenge (Oct 2025) — national university competition for UX design and innovation
- Secretary Treasurer, SOFEA Society (2022–2023); organized AI workshops for 200+ students